

Arachne + Mortal Blast

Revised baseline: SKeyDB level 70 and level 90. This replaces the earlier arbitrary-stat example.

Main sequence: **[Arachne Strike -> Mouchette Mortal Blast] x 5**. Ten played cards, four awakened Dramatic Encounter follow-ups, one enemy. Results are conditional non-critical Active damage before HP resolution.

Both characters at	Arachne Strike total	Mortal Blast total	Follow-ups total	Combined total
70	2,180	6,942	7,638	16,760
90	2,615	8,320	9,153	20,088

Real stat inputs, explicit baseline conditions

We use SKeyDB's published stat growth and skill coefficients. Both characters have max Gnostic Potential (+10 base-stat levels), skill level 6, no equipment and no active seasonal talent. Starting STR is zero. These choices define an unequipped reference baseline; they do not claim to reproduce a geared endgame team. [S1-S4]

Mouchette is E2, roused, and has already used her level-6 Exalt. E3 and Absolute Axiom are absent: Mortal Blast has two hits, the roused follow-up has three, and the cap is four follow-ups. Arachne is unroused; her conditional attached action/Fate Cut is inactive. All hits are explicitly non-critical, not an average over crit RNG. [S3-S5]

There are no starting follow-up stacks or partial pair progress. Sufficient cards and energy are supplied to perform all five pairs. Mortal Blast can copy the Arachne Strike; that copy normally remains owned by Arachne. The final generated copy is not played and adds no damage. Five Mortal Blasts must be available; this sequence does not generate those five casts by itself. [S3, P2]

If you meant only five Mortal Blasts

With Arachne present but no Arachne Strike played, two follow-ups trigger under the same zero-progress starting condition: **8,682 at level 70** or **10,400 at level 90**. The all-five-pairs interpretation above includes five extra played Strikes and two extra follow-ups.

No client answer or replay total was used to obtain these numbers. Our JavaScript pipeline calculates them from the sourced inputs. The Exalt's initial attack is outside the requested five-pair window and is not included.

Where every input comes from

SKeyDB snapshot: a2ff07765b2e0c78af827577f881054dbe224d8d. Retrieved 19 September 2026.

Character attack

SKeyDB's primary-stat formula is **ceil((20 + character level + Gnostic bonus levels) x growth - 0.000000001)**. Its catalog gives ATK growth 1.65 for Mouchette and 1.15 for Arachne, with default Gnostic bonus levels 10. We use no Soulforge percentage or equipment additions. [S1, S2]

Character	Level 70	Level 90
Mouchette	$\text{ceil}((20 + 70 + 10) \times 1.65) = 165$	$\text{ceil}((20 + 90 + 10) \times 1.65) = 198$
Arachne	$\text{ceil}((20 + 70 + 10) \times 1.15) = 115$	$\text{ceil}((20 + 90 + 10) \times 1.15) = 138$

Level-6 skill inputs

Event / contribution	SKeyDB coefficient	Level 70 input	Level 90 input
Arachne Strike	20% of Arachne ATK	$115 \times 0.20 = 23$	$138 \times 0.20 = 27.6$
Mortal Blast, per hit	30% of Mouchette ATK	$165 \times 0.30 = 49.5$	$198 \times 0.30 = 59.4$
Dramatic Encounter, per hit	30% of Mouchette ATK	49.5	59.4
Exalt flat Strike buff	250% of Mouchette ATK	$\text{ceil}(165 \times 2.50) = 413$	$\text{ceil}(198 \times 2.50) = 495$

The raw damage arguments retain their decimals until the pipeline rounding stage. The Exalt buff is different: our recovered state-initialization path rounds its property contribution upward when the buff is applied. Thus 412.5 becomes a stored +413, before later damage calculations. [P1, P3]

No invented positive STR or damage bonuses

Starting STR is zero, not 100. All other offensive and target modifiers are neutral for this reference case. Crit chance still exists on the characters, but we condition on non-critical outcomes; this is not an expected-damage calculation. The equipped result needs actual wheel, covenant, posse, relic and battle-state inputs.

Run the recovered pipeline

Flat bonuses, personal stacks and per-hit rounding explain the total.

Arachne stays separate from Mouchette's personal stacks

Arachne's non-critical pre-hit amount is **ceil(raw Strike + Exalt flat bonus)** under neutral modifiers: level 70 gives $\text{ceil}(23 + 413) = 436$; level 90 gives $\text{ceil}(27.6 + 495) = 523$. Mouchette's personal inside-Strike stack does not transfer to her. [P1-P3]

Mortal Blast and Dramatic Encounter

At skill level 6, Mortal Blast and the follow-up happen to have the same raw coefficient here. Their per-hit subtotal is $49.5 + 413 = 462.5$ at level 70, or $59.4 + 495 = 554.4$ at level 90. Mortal Blast deals two hits and the awakened follow-up deals three. [S3-S5]

With k previously completed follow-ups, multiply that subtotal by $1 + 0.25k$. The original offensive pipeline ceilings after subtracting 0.00001, then floors after neutral reductions with +0.00001. The neutral target step ceilings again. In this example this gives the per-hit values below. [P1]

Prior layers k	Personal factor	Level 70, per hit	Level 90, per hit
0	1.00x	463	555
1	1.25x	579	693
2	1.50x	694	832
3	1.75x	810	971
4	2.00x	925	1109

Example: at level 70 with one prior layer, $462.5 \times 1.25 = 578.125 \rightarrow$ **579 per hit**. Mortal Blast gives $2 \times 579 = 1,158$; its following Dramatic Encounter gives $3 \times 579 = 1,737$. Round each hit before summing. Do not round only the whole card total.

Our assumed schedule resolves one follow-up after each of the first four card pairs. The new +25-point layer is awarded after its damage. Pair five uses the existing 2.00x factor but triggers no follow-up because the cap is reached. This ordering uses prior code evidence; the complete event sequence is not gameplay-verified. [P2]

All five pairs, both levels

Each row includes one Arachne Strike and one Mortal Blast. Follow-up damage is included where the cap permits it.

Both characters level 70

Pair	Arachne	Blast: 2 hits	Follow-up: 3 hits	Total
1	436	926	1389	2751
2	436	1158	1737	3331
3	436	1388	2082	3906
4	436	1620	2430	4486
5	436	1850	Cap reached	2286

2,180 + 6,942 + 7,638 = 16,760 Active damage before HP resolution.

Both characters level 90

Pair	Arachne	Blast: 2 hits	Follow-up: 3 hits	Total
1	523	1110	1665	3298
2	523	1386	2079	3988
3	523	1664	2496	4683
4	523	1942	2913	5378
5	523	2218	Cap reached	2741

2,615 + 8,320 + 9,153 = 20,088 Active damage before HP resolution.

Five Mortal Blasts without playing Arachne

Factors are 1.00, 1.00, 1.25, 1.25, 1.50; follow-ups occur after casts 2 and 4. Level 70: Blast 5,554 + follow-ups 3,128 = 8,682. Level 90: Blast 6,650 + follow-ups 3,750 = 10,400. The values below are generated by the same pipeline, not scaled from the combined total.

Scope, evidence and next steps

A sourced character baseline is more useful than arbitrary inputs, but it still needs a fully specified battle to predict actual HP loss.

These are one-target, unmitigated, non-critical reference results. Enemy shields, vulnerability, reduction, caps, death, seasonal buffs and random target allocation can change them. Arachne's enabled attached action can add prism and Fate Cut effects between cards; the inactive case used here excludes those events. Our Frenzy replay is a separate, more complex scenario.

The SKeyDB snapshot is an external data source, not proof that its current region/version exactly matches PC144/build51. Its coefficients and described follow-up mechanics used here agree with our prior recovered expressions. A complete baseline battle has not been independently checked. We still have zero completed gameplay predictions and zero holdouts.

Source links and attribution

S1: [Stat growth and defaults - SKeyDB](#)

S2: [Primary-stat scaling implementation - SKeyDB](#)

S3: [Mortal Blast coefficients and E2 copy - SKeyDB](#)

S4: [Exalt coefficients and E3 distinction - SKeyDB](#)

S5: [Rouse and follow-up changes - SKeyDB](#)

S6: [Arachne Strike coefficients - SKeyDB](#)

S7: [Dramatic Encounter coefficient - SKeyDB](#)

SKeyDB by dansa and contributors supplies the cited dataset and scaling reference. Its original data/content is licensed CC BY-NC-SA 4.0; this adapted baseline write-up is shared under the same terms. Game-owned material remains owned by its respective rights holders. No artwork or extracted client files are included.

P1: [docs/DAMAGE_FORMULA.md](#); [engine/show-damage.mjs](#); [engine/active-target.mjs](#)

P2: [docs/MOUCHETTE.md](#); [docs/CARD_ATTRIBUTION.md](#)

P3: [docs/STATE_PROPERTY_ROUTING.md](#): buff routing and initialization ceiling

Reproduction

Run `node tools/mortal-blast-example.mjs`. It reads the pinned SKeyDB inputs and writes `output/pdf/mortal-blast-example-calculation.json` with all input values and hit traces. The earlier 57,478 result used arbitrary stats and is superseded by this revision. This example adds no gameplay verification credit.